QND Computer Science Day 1

Mr. Schmidt

DOWNLOAD SWIFT PLAYGROUNDS NOW

Agenda

- Expectations
- Programming Languages
- Swift
- Tools
- Your first (actual) program

Expectations

- You will not master everything today
- You will (probably) be lost
- That's okay!

Expectations

- Practice, practice, practice
- Help each other!
- Today will be slower and more boring than normal
- Bear with me!

Machine Code

- Hard to write and understand
- Different computer types use different machine code

```
.text:00000000 _sub:     push    ebp
.text:00000001           mov     ebp, esp
.text:00000003           mov     eax, [ebp+8]
.text:00000006           mov     ecx, [ebp+0Ch]
.text:00000009           lea     eax, [ecx+eax*2]
.text:0000000C           pop     ebp
.text:0000000D           retn
.text:00000010 _main:    push    ebp
.text:00000011           mov     ebp, esp
.text:00000013           push    ecx
.text:00000014           mov     eax, [ebp+0Ch]
.text:00000017           mov     ecx, [eax+4]
.text:0000001A           push    ecx
.text:0000001B           call    dword ptr ds:__imp__atoi
.text:00000021           add     esp, 4
.text:00000024           mov     [ebp-4], eax
.text:00000027           mov     edx, [ebp-4]
.text:0000002A           push    edx
.text:0000002B           mov     eax, [ebp+8]
.text:0000002E           push    eax
.text:0000002F           call    _sub
.text:00000034           add     esp, 8
.text:00000037           mov     esp, ebp
.text:00000039           pop     ebp
.text:0000003A           retn
```

Programming Languages

- *Compile* to machine code
- Many, many different languages

S	
A	
B	
C	
D	
F	

TIERMAKER

Swift

- Programming Language created by Apple
- Used for creating iOS and macOS applications
- Simple and modern!

My Website

- Everything you need is at markschmidt.io/qnd
- Presentation slides
- Code links

Swift Playgrounds

- We need a coding environment
- We'll use Swift Playgrounds
- Normally for iOS App Development
- Use a template to do simple starter coding

Setup

- Go to markschmidt.io/qnd
- Click the "Download Template" button
- Open the file in the Files app
- Unzip
- Open the Playground

Why is this so complicated?

- Apple is dumb
- I am very mad

Robots!

- Move a robot around a virtual world
- Reach the end gate
- 8 different levels

Robot Commands

- robot.forward()
- robot.turnRight()
- robot.turnLeft()
- robot.wait()
- Don't touch a wall or it's game over!

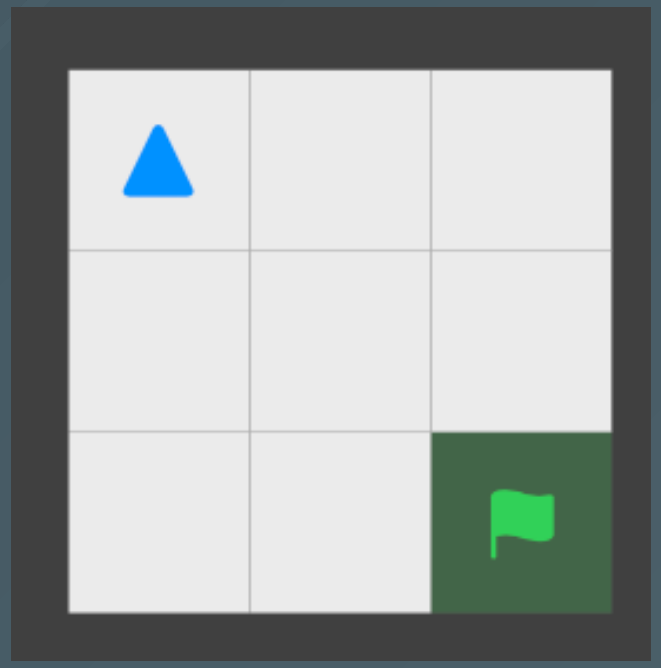
What does all this mean?

- `robot` is a reference to the robot on the screen
- `.` tells the program to apply the function to the robot
- `forward`, `turnRight`, etc. are the names of functions that tell the robot what to do
- `()` means to execute the function
  - Later, we'll see functions with inputs, which will go inside the parentheses

Level 1

- Turn right
- Move forward twice
- Turn right
- Move forward twice

```
robot.turnRight()
robot.forward()
robot.forward()
robot.turnRight()
robot.forward()
robot.forward()
```



Level 2

- More complicated movement!

Level 3

- Keys!
- Go get the key, then go through the lock to the end

Level 4

- Enemies!
- Avoid the enemies by waiting for them to pass
- Enemies move when you move
- If you touch them, game over
- Use `robot.wait()` to wait for an enemy to pass

Level 5

- Lasers!
- Periodically turn on and off

Level 6

- Teleporters
- Go through the teleporter to get to the end

Level 7

- More lasers!

Level 8

- Putting it all together